AI251P Applied Computer Vision

# Capstone Project – Phase 1 Report

Camera-Trap Image Classification

Project 5 · Phase 1 (M1–M3, 40 points) · Due 28 October 2026

## 1 Team and project

| Field | Your answer |
|---|---|
| Team name | Project 5 |
| Topic | Project 5: camera-trap image classification |
| Application area | Wildlife monitoring and image triage |
| Code repository | https://github.com/salehdev3/Capstone_5 |
| W&B project | https://wandb.ai/saleh-alanazi-ministry-of-interior/camera-trap-phase1 |
| Dataset card | https://github.com/salehdev3/Capstone_5/blob/v1/DATASET_CARD.md |

### Team members

| Name | Student ID | Workstream (Table 3) | Phase 1 contribution |
|---|---|---|---|
| Saud Abdullah Alfaiz | 225462 | Data and classification | Data preparation, grouped split, audit and card |
| Saleh Mohammed Alanazi | 225486 | Data and classification | Scratch CNN and evaluation |
| Yousef Nafi Alanazi | 225457 | Data and classification | Frozen and full ResNet-50 comparison |
| Mohannad Mohammed Alqahtani | 225461 | Integration | Independent review, comparison and report |

### Problem and acceptance criteria

- **Summary:** The system classifies full camera-trap frames as empty or animal, then predicts animal species on human-box crops. It reports per-class metrics, calibration and errors.
- **Input → output:** A camera-trap photograph → empty/animal decision; an annotated animal crop → species category.
- **Intended users and the decision supported:** Wildlife-data reviewers use it to prioritize animal frames and inspect species predictions.
- **Acceptance criteria (fixed before results):** Animal recall $\geq$ .99; empty removal $\geq$ .20; species macro-F1 $\geq$ .65; species ECE $\leq$ .10.

# 2 Data foundations (M1) (12 points)

## 2.1 Datasets

| Dataset | Link / ref. | Content and annotations | Use | Licence | Images |
|---|---|---|---|---|---|
| CCT20 subset | LILA CCT | Camera-trap frames and human boxes | All splits | CDLA-Permissive | 6,022 |
| Team-collected (bonus) | Private collection | Reviewed frames and boxes | Train/test | Team-owned; private | 19 |

The notebook selected 6,022 CCT20 images under a fixed seed of 42, then included 19 reviewed team images. This project-specific subset does not use the official CCT benchmark split.

## 2.2 Cleaning

| Check | Method / tool | Images removed |
|---|---|---|
| Corrupt or unreadable files | Download/decode checks | 0 |
| Exact duplicates | ID/hash checks | 0 |
| Near-duplicates | pHash cross-partition checks | 0 |
| Other (human audit quarantine) | One flagged event, four frames | 4 |
| **Images remaining** | 6,041 initially kept; 6,037 after audit | 6,037 |

## 2.3 Leakage-safe split

- **Grouping key:** CCT20 camera/event; local images by city and session.
- **Why this prevents leakage:** Frames from the same camera or burst share backgrounds, lighting and animals; grouping keeps these correlations within one partition.
- **Stratification and seed:** Group-aware approximate joint-label stratification, seed 42. Camera separation takes priority over exact class proportions.

| Split | No. of Groups | No. of Images | Images per class (or % of the smallest class) |
|---|---|---|---|
| Train | 13 | 3,534 | Empty: 319; animal: 3,215 |
| Validation selection | 3 | 1,337 | Empty: 91; animal: 1,246 |
| Validation calibration | 2 | 527 | Empty: 13; animal: 514 |
| Test | 4 | 639 | Empty: 53; animal: 586 |

## 2.4 Label audit

| Audited sample size | Errors found | Estimated label-error rate | Upper bound (95% confidence) |
|---|---|---|---|
| 10 training events | 1 | 10% | 39.4% one-sided |

One frame was drawn from each of 10 random training events; the team found one incorrect label and quarantined the affected event. A separate 20-image review signed by Mohannad Mohammed Alqahtani found no additional frame-label errors; five box checks were not applicable.

### 2.5 Data loader and augmentation

RGB images were prepared with maximum side 1,024 pixels, resized to 224×224, then ImageNet-normalized. Training alone uses random flip and brightness/contrast augmentation from the shared preprocessing module; these variations reflect camera viewpoint and illumination. Exact augmentation parameters are fixed in `preprocessing.py`.

### 2.6 Dataset card summary

Raccoon and opossum dominate some groups while rare species have few examples; camera-group splits have distribution shift. CCT20 is largely US wildlife, and five held-out Tabuk frames cannot establish Saudi deployment performance. The small audit has a wide error bound. Source terms and course policy restrict image redistribution; no GPS or private photos are published. The full card is linked in Section 1.

## 3 Classification (M2–M3) (20 points)

**Task:** Full frames are classified empty/animal; human-box crops are classified by species. These classifiers provide the baseline for later detection and segmentation.

### 3.1 Runs compared

Six regimes were compared using validation-selection macro-F1; frozen ResNet-50 was chosen for both tasks. The locked test set was evaluated after selection. W&B links for all six runs are in the linked project.

| Run (W&B link) | Backbone | Training | Input res. | Epochs | Val. acc. | Val. macro-F1 |
|---|---|---|---|---|---|---|
| Triage (W&B) | CNN | From scratch | 224 | 6 | .941 | .601 |
| Triage (W&B) | ResNet-50 | Pretrained, frozen | 224 | 12 | .940 | .662 |
| Triage (W&B) | ResNet-50 | Pretrained, full fine-tuning | 224 | 12 | .927 | .640 |
| Species (W&B) | CNN | From scratch | 224 | 12 | .505 | .156 |
| Species (W&B) | ResNet-50 | Pretrained, frozen | 224 | 12 | .761 | .364 |
| Species (W&B) | ResNet-50 | Pretrained, full fine-tuning | 224 | 12 | .750 | .340 |

**Test-set results (final models only)**

| Run | Test accuracy (95% CI) | Test macro-F1 | ECE |
|---|---|---|---|
| Triage scratch | .917 [.868,.966] | .478 | .0463 |
| Triage frozen (selected) | .933 [.897,.967] | .652 | .0460 |
| Species scratch | .350 [.284,.386] | .178 | .0733 |
| Species frozen (selected) | .742 [.642,.876] | .482 | .0568 |

**Which model you keep and why:** Frozen ResNet-50 gave the highest validation macro-F1 for both tasks (.662 triage and .364 species). It was therefore retained before test results were examined. At test threshold .3416, triage retained 586/586 animal frames but removed only 6/53 empty frames (11.3%), missing the 20% empty-removal target. Species test macro-F1 was .482, below the .65 target; ECE .0568 met the .10 limit.

### 3.2 Learning curves

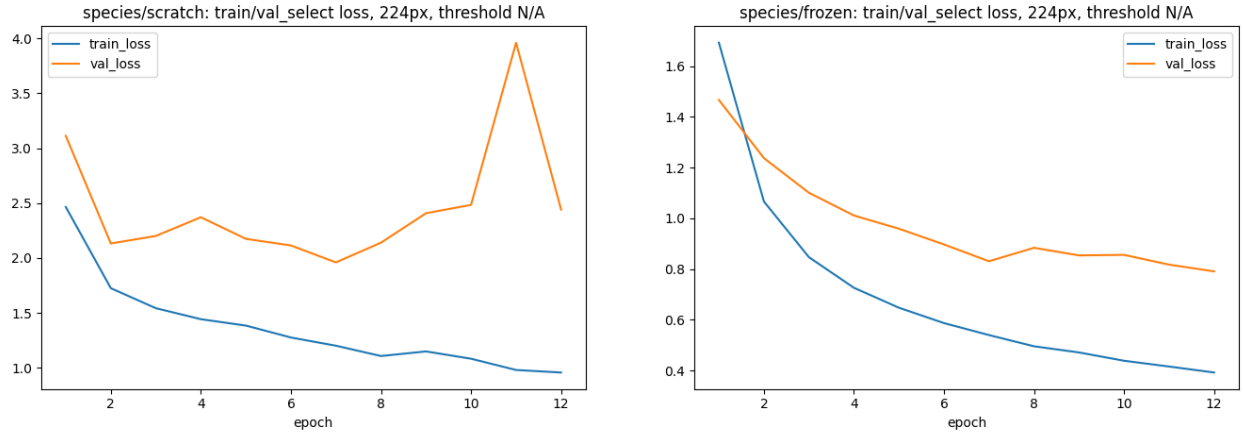



Figure 1: Training and validation-selection loss: (left) scratch CNN, (right) frozen ResNet-50. Split: train and validation-selection; input: 224×224; threshold: argmax.

Species full fine-tuning reached 94.7% training accuracy but 75.0% validation accuracy, consistent with overfitting. Frozen species reached 89.7% and 76.1%, respectively. Rare-class support and camera shift remain important limits.

### 3.3 Per-class results (selected model, test set)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Empty (triage) | .917 | .208 | .338 | 53 |
| Animal (triage) | .933 | .998 | .965 | 586 |
| Badger (species) | .000 | .000 | .000 | 1 |
| Bird (species) | .900 | 1.000 | .947 | 9 |
| Bobcat (species) | .629 | .684 | .655 | 57 |
| Cat (species) | .632 | .250 | .358 | 48 |
| Coyote (species) | .956 | .763 | .849 | 169 |
| Deer (species) | .000 | .000 | .000 | 3 |
| Dog (species) | .676 | .862 | .758 | 29 |
| Fox (species) | .000 | .000 | .000 | 1 |
| Opossum (species) | .656 | .966 | .781 | 87 |
| Rabbit (species) | .250 | .250 | .250 | 4 |
| Raccoon (species) | .366 | .536 | .435 | 28 |
| Rodent (species) | .000 | .000 | .000 | 0 |
| Skunk (species) | .933 | .737 | .824 | 19 |
| Squirrel (species) | .860 | .925 | .891 | 53 |
| **Macro average (species)** | .490 | .498 | .482 | 508 |

Top-$k$ accuracy (species, $k = 3$): .9055 on test; 224×224, argmax.

### 3.4 Confusion matrix and calibration

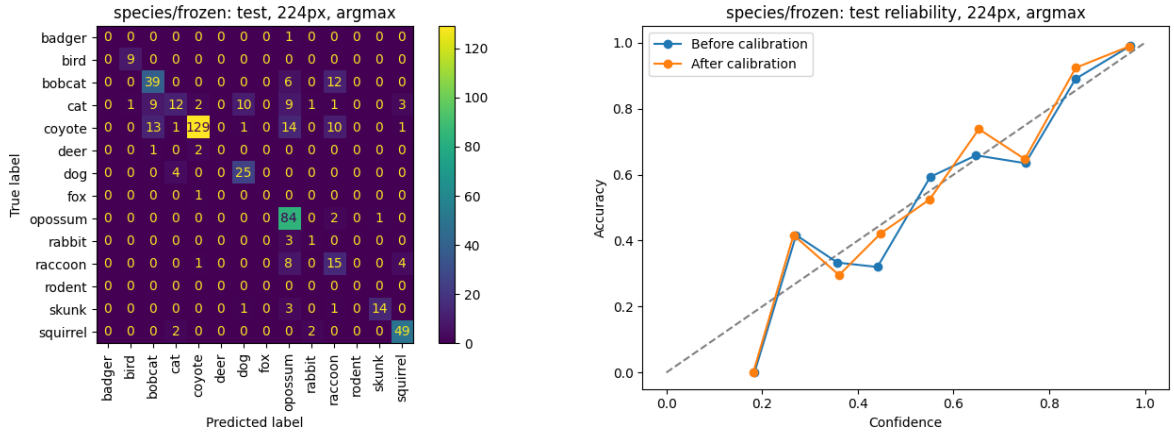



Figure 2: (Left) confusion matrix; (right) reliability diagram. Split: test; input: 224×224; threshold: argmax.

Bobcat was repeatedly confused with raccoon in the gallery; a fox crop was called coyote. Selected species test ECE was .0568 versus raw .0454, so temperature scaling did not improve ECE on this test split. Gallery observations are not a causal proof.

### 3.5 Error analysis

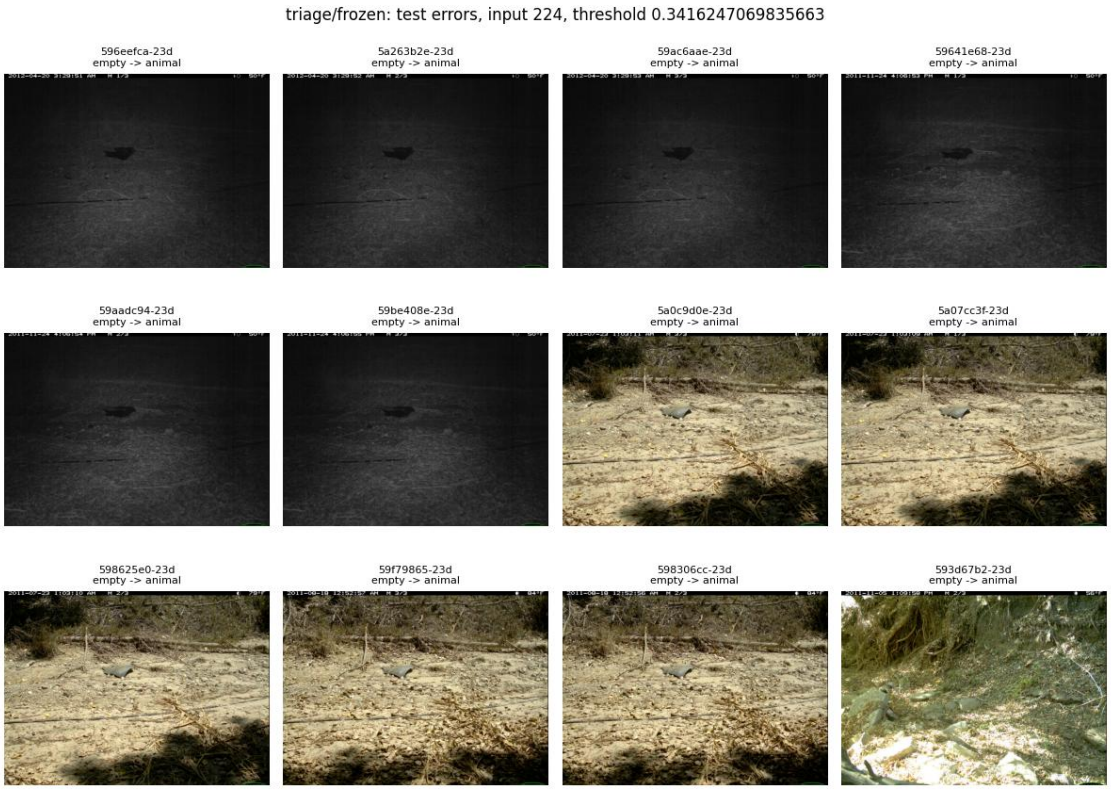



Figure 3: Triage test error gallery: recorded label and prediction are shown; input 224×224, animal threshold .3416. Several source labels need human verification.

| Error pattern | Examples | Likely cause (data, label, model) |
|---|---|---|
| Recorded empty → animal | 12 shown | Several daylight frames look bird-like; source-label review needed. |
| Bobcat → raccoon | 5 shown | Low-detail night crops and camera-domain shift are plausible contributors. |

### 3.6 Data-improvement plan

| Finding from the error analysis | Planned action | Expected effect |
|---|---|---|
| Possible source-label errors | Review full-resolution originals; report corrected-label sensitivity separately | Distinguish annotation issues from model errors |
| Rare species and empty frames | Add verified, independently grouped examples; tune on validation only | Improve minority-class recall and empty removal |

## 4 Reproducibility (8 points)

| Item | Your answer |
|---|---|
| Repository and commit hash | https://github.com/salehdev3/Capstone_5, evaluated results release `2cc1731` on `v1` |
| Configuration file used for the reported runs | `handoff/data_config.json` and `runs/member2,member3/config.json` |
| Random seeds | 42 |
| Requirements file (pinned versions) | Role-specific `handoff/requirements_member*.txt` and runtime JSON records |
| Command(s) to reproduce the main result | Run `notebooks/Phase1_Final_Presentation.ipynb` in section order on Colab GPU |
| Where the data is obtained (not committed) | CCT20 on LILA; private team photos; recorded manifest hash in dataset card |
| Reproduced by (team member) and result | Fresh-process inference recorded in `review/fresh_process_results.json`; complete clean replay pending |

## 5 Submission checklist

- ☐ Corrupt files and duplicates handled, with counts reported
- ☐ Split is grouped by the source of correlation; the grouping key is justified
- ☐ Label audit reported with a confidence bound
- ☐ Dataset card states licences, limitations and privacy measures
- ☐ Scratch and fine-tuned classifiers compared in W&B
- ☐ Per-class metrics, confusion matrix and calibration (ECE) reported
- ☐ Test set used only for final evaluation, not model selection
- ☐ Data-improvement plan follows from the error analysis
- ☐ Fixed seeds, single configuration, pinned requirements; no hard-coded paths
- ☐ Every figure states split, input resolution and thresholds
- ☐ No data or personal information committed to the repository

## 6 Self-assessment

| Criterion | Points | Self-score | Strongest evidence (section) | Known gap |
|---|---|---|---|---|
| Data foundations and dataset card (M1) | 12 | 10 | Cleaning, split, audit, card | Small initial audit |
| Classification (M2–M3) | 20 | 15 | Six runs, metrics and galleries | Two preset targets missed |
| Reproducibility | 8 | 6 | Config, pinned requirements, hashes | Full clean replay pending |
| **Total** | **40** | **31** | Conservative team estimate | Instructor decides final score |

## 7 Obstacles and how they were addressed

Colab upload stalls interrupted evaluation; results are now saved before bounded W&B upload. Rare classes and uncertain source annotations limit classification. The original locked-test results are retained while suspicious examples await separate human adjudication.

| Obstacle | Module | Impact on the work | Action taken | Status |
|---|---|---|---|---|
| Colab upload stalls | M2–M3 | Evaluation interruption | Save metrics before bounded W&B upload | Resolved |
| Rare classes | M1–M3 | Low species macro-F1 | Report support and plan more verified data | Open |
| Source-label uncertainty | M1–M3 | Some apparent test errors | Independent review and separate adjudication | Open |